

Introduction to Meta-Analysis

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Import libraries

```
library(dplyr)
library(meta)
library(readxl)
```

Load data

```
AcuteManiaP <- AcuteManiaP_raw %>% filter(treat2 == "Risperidone")

str(AcuteManiaP)

## tibble [5 x 15] (S3: tbl_df/tbl/data.frame)
##  $ studlab: num [1:5] 23 24 25 56 58
##  $ treat1 : chr [1:5] "Placebo" "Placebo" "Placebo" "Placebo" ...
##  $ treat2 : chr [1:5] "Risperidone" "Risperidone" "Risperidone" "Risperidone" ...
##  $ ncont1 : num [1:5] 142 72 119 47 138
##  $ mean1  : num [1:5] -10.5 -10.3 -4.8 -8.2 22.1
##  $ sd1    : num [1:5] 15.5 11.9 9.5 10 10.2
##  $ ncont2 : num [1:5] 144 68 127 51 153
##  $ mean2  : num [1:5] -22.7 -14.5 -10.6 -14.3 17
##  $ sd2    : num [1:5] 13.2 12.4 9.5 10 10.2
##  $ event1 : num [1:5] 51 30 29 NA 39
##  $ n1     : num [1:5] 145 76 125 51 140
##  $ event2 : num [1:5] 105 40 55 NA 65
##  $ n2     : num [1:5] 146 75 134 52 154
##  $ AE1    : num [1:5] 5 1 4 0 5
##  $ AE2    : num [1:5] 3 0 0 0 2

nrow(AcuteManiaP)

## [1] 5
```

Meta-analysis of a continuous outcome

```
pooledSMD1 <- metacont(mean.c = mean1, sd.c = sd1, n.c = ncont1,
                        mean.e = mean2, sd.e = sd2, n.e = ncont2,
                        data = AcuteManiaP, studlab = studlab, sm = "SMD")

summary(pooledSMD1)
```

```
##           SMD           95% CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      23.8
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.5
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.5
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.6
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.6
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##           SMD           95% CI      z  p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5943 [-0.7627; -0.4259] -6.92 < 0.0001
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0158 [0.0000; 0.2555]; tau = 0.1256 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 6.91   4  0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

5 studies were involved in the meta-analysis the fixed effect model gave an estimate of -0.60 [-0.73; -0.48] for the fixed effects and -0.59 [-0.76; -0.43] for the random effects. Both significant.

The test of heterogeneity does not highlight evidence of heterogeneity. This doesn't necessarily mean that there's no heterogeneity, it might be due to low power, i.e. a small number of study.

```
summary(update(pooledSMD1, method.tau = "PM"))
```

```
##           SMD           95% CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      24.0
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.2
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.6
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.2
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.9
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##           SMD           95% CI      z  p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5952 [-0.7579; -0.4325] -7.17 < 0.0001
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0134 [0.0000; 0.2555]; tau = 0.1160 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
## Test of heterogeneity:
```

```
##      Q d.f. p-value
## 6.91    4 0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Paule-Mandel estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

The random effects model now estimates that the treatment effect is -0.60 [-0.76; -0.43]

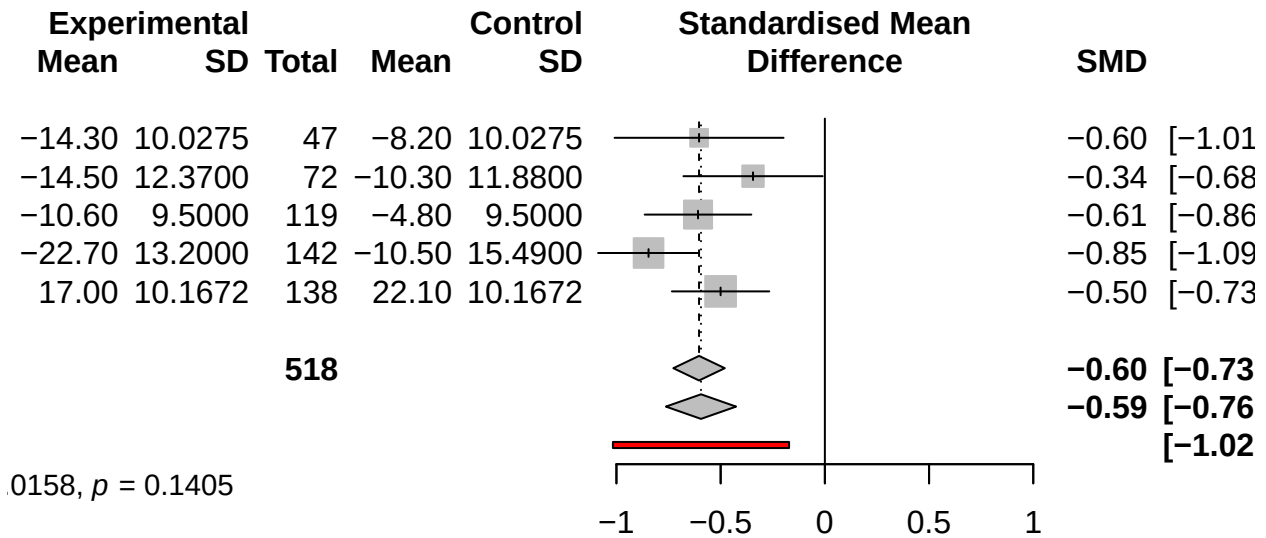
However, with only 5 studies, it's difficult to estimate the heterogeneity (we need at least >5 studies).

```
summary(update(pooledSMD1, method.random.ci = T))
```

```
##      SMD      95% CI %W(common) %W(random)
## 23 -0.8460 [-1.0881; -0.6039]      26.0      23.8
## 24 -0.3446 [-0.6786; -0.0107]      13.6      16.5
## 25 -0.6086 [-0.8645; -0.3528]      23.3      22.5
## 56 -0.6036 [-1.0091; -0.1980]       9.3      12.6
## 58 -0.5003 [-0.7340; -0.2666]      27.9      24.6
##
## Number of studies: k = 5
## Number of observations: o = 1061 (o.e = 543, o.c = 518)
##
##      SMD      95% CI  z|t  p-value
## Common effect model -0.6036 [-0.7270; -0.4802] -9.59 < 0.0001
## Random effects model -0.5943 [-0.8249; -0.3638] -7.16  0.0020
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0158 [0.0000; 0.2555]; tau = 0.1256 [0.0000; 0.5054]
## I^2 = 42.1% [0.0%; 78.7%]; H = 1.31 [1.00; 2.17]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 6.91    4 0.1405
##
## Details of meta-analysis methods:
## - Inverse variance method
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
## - Hartung-Knapp adjustment for random effects model (df = 4)
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

Forest plots

```
forest(pooledSMD1, sortvar = ncont1 + ncont2,
       prediction = T, common = T)
```



Meta-analysis of a binary outcome

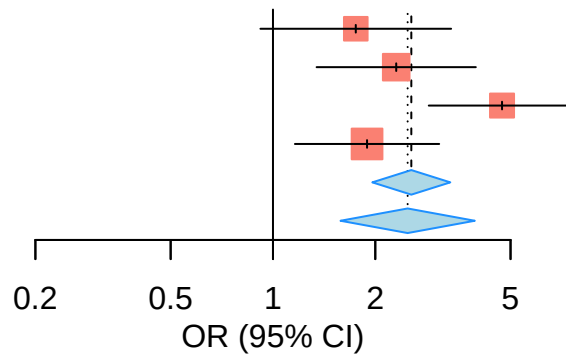
Odds Ratio as Effect Size

```
pooledOR <- metabin(event.e = event2,
  n.e = n2,
  event.c = event1,
  n.c = n1,
  data = AcuteManiaP,
  studlab = studlab, sm = "OR")
summary(pooledOR)
```

```
##          OR          95% CI %W(common) %W(random)
## 23 4.7202 [2.8731; 7.7548]    20.7      26.5
## 24 1.7524 [0.9185; 3.3434]    20.0      21.7
## 25 2.3047 [1.3439; 3.9524]    25.4      25.1
## 56      NA                0.0        0.0
## 58 1.8914 [1.1603; 3.0832]    33.9      26.7
##
## Number of studies: k = 4
## Number of observations: o = 1098 (o.e = 561, o.c = 537)
## Number of events: e = 414
##
##          OR          95% CI      z  p-value
## Common effect model 2.5529 [1.9603; 3.3248] 6.95 < 0.0001
## Random effects model 2.4903 [1.5804; 3.9242] 3.93 < 0.0001
```

```
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.1391 [0.0000; 2.7687]; tau = 0.3730 [0.0000; 1.6640]
## I^2 = 65.8% [0.0%; 88.4%]; H = 1.71 [1.00; 2.93]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 8.78    3 0.0324
##
## Details of meta-analysis methods:
## - Mantel-Haenszel method (common effect model)
## - Inverse variance method (random effects model)
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q
forest(pooledOR, sortvar = n1 + n2, layout = "JAMA", allstudies = F,
      fs.study = 10, ff.study = "italic",
      col.diamond = "indianred", col.diamond.lines = "dodgerblue",
      col.square = "salmon", col.square.lines = "salmon")
```

Source	OR (95% CI)
24	1.75 [0.92; 3.34]
25	2.30 [1.34; 3.95]
23	4.72 [2.87; 7.75]
58	1.89 [1.16; 3.08]
Total (common effect)	2.55 [1.96; 3.32]
Total (random effect)	2.49 [1.58; 3.92]



Heterogeneity: $\chi^2_3 = 8.78$ ($P = .03$), $I^2 = 65.8\%$

Risk Ratio as Effect Size

```
summary(update(pooledOR, sm = "RR"))

##      RR      95% CI %W(common) %W(random)
## 23 2.0447 [1.6034; 2.6075]      33.7      34.1
## 24 1.3511 [0.9524; 1.9168]      19.6      21.9
```

```

## 25 1.7692 [1.2122; 2.5820]      19.8      19.6
## 56      NA                      0.0      0.0
## 58 1.5152 [1.0954; 2.0957]      26.9      24.3
##
## Number of studies: k = 4
## Number of observations: o = 1098 (o.e = 561, o.c = 537)
## Number of events: e = 414
##
##              RR          95% CI    z  p-value
## Common effect model  1.7116 [1.4648; 2.0001] 6.76 < 0.0001
## Random effects model 1.6874 [1.3858; 2.0546] 5.21 < 0.0001
##
## Quantifying heterogeneity (with 95% CIs):
## tau^2 = 0.0142 [0.0000; 0.4305]; tau = 0.1191 [0.0000; 0.6562]
## I^2 = 31.5% [0.0%; 75.4%]; H = 1.21 [1.00; 2.02]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 4.38   3 0.2231
##
## Details of meta-analysis methods:
## - Mantel-Haenszel method (common effect model)
## - Inverse variance method (random effects model)
## - Restricted maximum-likelihood estimator for tau^2
## - Q-Profile method for confidence interval of tau^2 and tau
## - Calculation of I^2 based on Q

```